Chapter 7
Microbial Consortia for Effective Degradation and Decolorization of Textile Effluents

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7.1 Introduction

Dyes perhaps are defined as substances that when applied to substrates provide color by a process that alters, at least temporarily, any crystal structure of the colored substances. It has been estimated that more than 10,000 different types of dyes and pigments are used in the industries and over 7×10^5 metric tons per year of synthetic dyes are produced globally. The most common areas where these synthetic dyes are widely employed are textile, food, pharmaceutical, plastics, cosmetics, and photographic and paper industries. In India alone, the dyestuff industry produces around 60,000 metric tons of dyes, which is approximately 6.6% of total colorants used worldwide. The largest consumer of the dyes is the textile industry accounting for two-thirds of the total production of dyes.

Generally in textile industry, continuous, semi-continuous, or batch processes are the methods for dyeing textile materials. Among these, batch process is the most common method and produces large volumes of effluents, often rich in color, and chemicals and also generates waste as sludge, fibers, and chemically polluted waters. This effluent requires proper treatment before being discharged into the natural resources. If these effluents are discharged into the natural resources without proper treatment, they degrade the quality of the soil and water and its dependent habitats and environment also. Hence, these industries are now facing major problems in environment pollution. To mitigate this problem, industries and researchers are focusing on the reduction of textile wastewater and the formulation of alternative efficient treatment techniques.

In general, the dye removal methods have been classified into three principal categories: physical, chemical, and biological methods. The physical methods

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include precipitation (coagulation, flocculation, and sedimentation), adsorption (on activated carbon and biological sludge), filtration, or reverse osmosis membrane processes. Unfortunately, most of the dyes used in textile industry escape these conventional effluent treatment processes and persist in the environment, as a result of their high stability to light, temperature, detergents, chemicals, soap, and other parameters such as bleach and perspiration. Moreover, some limitations in this process can be pointed out, such as cost, low efficiency, labor-intensive operation, and lack of selectivity of the process. These limitations made scientific community in order to search for cost-effective, environmental-friendly alternative methods for effective conversion of toxic textile dyes into nontoxic compounds.

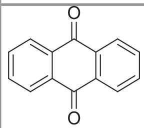
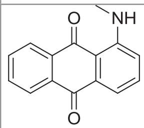
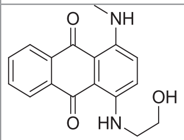
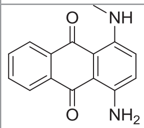
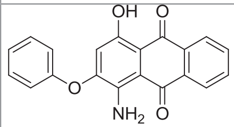
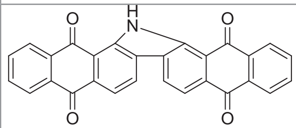
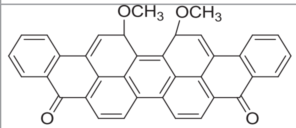
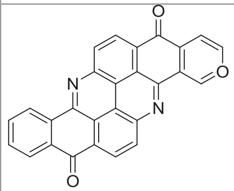
A number of biotechnological approaches have been suggested by researchers as of potential interest in combating with this problem in an eco-efficient manner. Microorganisms are more capable of dye decolorization in general; bacteria either in pure cultures or in consortia in particular have been reported. The applications of microbial consortia in synthetic dye degradation process offer considerable advantages over the use of pure cultures (Yadav et al. 2020c, d). Different strains may interact with dye molecule at different positions. Decomposition of complex organic materials to simple produced by one bacterium, further these simple components may use by another bacterial strains. Therefore, bacterial consortium was developed using selected (five) bacterial strains which are isolated from the textile effluents and also investigated the biodegradation efficiency of toxic anthraquinone dyes, i.e., Acid Blue 25 turned into environmentally nontoxic metabolic compounds which may be confirmed by toxicological analysis.

7.2 Structure and Classification of Textile Dyes

New dyes are continuously being developed to meet the demands of current technology, in fabrics, use of several advanced dyeing equipment, and to overcome the serious environmental concerns associated with some existing dyes. Generally, textile dyes are classified either in accordance with their chromophore group or in their application to textile fibers and other coloring applications. There are some commonly available chromophore groups, namely ethylenic, nitro, azo, keto, niroso, thioketo, quinonoids, and acridine. Among these, anthraquinone dyes come under quinonoid chromophore group.

Generally, anthraquinone dyes comprise two carbonyl groups on both the sides of a benzene ring (Table 7.1, A). Its dye variants are formed by changing the substituent (amino, hydroxyl, halogen, and sulfonic acid), and the properties and locations of the substituent affect the dye color. Electron-donating substituent and electron-accepting substituent have different effects, with the introduction, and therefore have little influence on color. In contrast, the electron-donating substituent (i.e., amino or substituted amino groups) has a significant impact on dyes with color changing from yellow to red, purple, or blue (Table 7.1, A–D).

Table 7.1 Basic structures of several Anthraquinone dyes

S.No	Name of the dye	CAS	Chemical structure
A	Anthraquinone (Yellow)	84-65-1	
B	1-(Methylamino)Anthraquinone (Red)	82-38-2	
C	1-[(2-Hydroxy amino]-4-(methylamino) Anthraquinone (Purple)	86722-66-9	
D	1-Amino-4-(methylamino) anthracene-9,10-dione (Blue)	1220-94-6	
E	Disperse Red-3B	84-65-1	
F	Vat Yellow-28	4229-15-6	
G	Vat Green-1	128-58-1	
H	Pigment Yellow-24	475-71-8	

Sources: Li et al. (2019)

Anthraquinone dyes are further divided into four categories as follows: (1) anthraquinone derivatives, such as Disperse Red 3B (Table 7.1, E) and Acid Green 25; Acid Blue 25; (2) heterocyclic anthraquinone dyes, such as Vat Yellow 28 (Table 7.1, F); (3) fused ring anthrone dyes, such as Vat Green 1 (Table 7.1, G); and (4) heterocyclic anthrone dyes, such as Pigment Yellow 24 (Table 7.1, H). Due to the presence of different substituents in anthraquinone dye, the color of the dye varies greatly, and this is the main reason why the same strains have different degradation efficiencies for different anthraquinone dyes.

7.3 Dye Degrading Microorganisms

Bacteria play far more important ecological role in natural environments with their small size which suggests, as an ecological perspective, these bacteria are the part of microbial communities. Each organism in a part of an ecosystem interacts with its surroundings and in some cases greatly modifies the profile characteristics of the environment with their biological process. This is particularly true with bacteria, where significant chemical changes can occur because of their vast metabolic activities.

Microbial ecologists are primarily interested in two aspects in microbiology of ecosystem concerned to bacteria, i.e., (i) the bacterial presence and (ii) the nature and extent of their metabolic activity. The knowledge of the qualitative and quantitative aspects of these two concerns and yields valuable information about the potential function in ecosystem for biogeochemical cycles. Just because the presence of an organism not only enough to overcome the problem necessarily, but they might be metabolically active. Since the only active organisms have an ecological impact. The measurable occurrence of a biochemical transformation thus serves as good evidence that a specific group of bacteria has to be present as well as it should metabolically active. Environmental factors affect the growth of bacteria in nature, as they do in culture. Thus, temperature, pH, water availability, and oxygen all play a role in governing the composition and number of bacteria in a particular habitat.

The microbes can degrade, assimilate, or absorb a wide range of textile dyes (Singh et al. 2020). The biological treatment is more effective than conventional modes of treatment, which is low cost and less accumulation, and produces harmless sludge. Hence, the biological treatment was an eco-friendly and less toxic and causes mineralization of dyes in an easy way. The majority of microbial enzymes currently used in industry are extracellular hydrolytic enzymes (Devi et al. 2020; Kour et al. 2019a). The main theme of decolorization and degradation of anthraquinone dyes was the breakdown of bonds, leading to removal of color, and also produces its metabolites. Anthraquinone dyes are known to undergo reductive cleavage, where aromatic amines are metabolized under aerobic conditions. These enzymes can be produced by a large number of microorganisms which include bacteria (*Aeromonas hydrophila* DN322, *Bacillus cereus* strain DC11, *Escherichia coli* DH5α, *Staphylococcus* sp. K 2204, and *Staphylococcus hominis* subsp. hominis DSM 20328), fungi (*Schizophyllum commune* IBL-06, *P. eryngii* F032), algae (*Lyngbya* sp. BDU 9001), and yeast strains (*C. guilliermondii* Y011, *C. dubliniensis* Y014, *C. guilliermondii* Y004, *C. famata* Y003, and *Kluyveromyces marxianus* IMB3) (Rana et al. 2019; Rastegari et al. 2019; Yadav 2020; Yadav et al. 2020a, b).

7.4 Enzymes Involved in Dye Degradation

There are two major steps that have been involved in degradation of anthraquinone dyes, which consist of cleavage of aromatic amine ring and mineralization of intermediates. The biodegradation ability of bacteria is associated with its intracellular and extracellular oxidoreductive enzyme system such as laccases, lignin peroxidase, manganese peroxidase, tyrosinases, and anthraquinone reductase. However, a single bacterial species may produce more than one enzyme that can actively participate in the biodegradation process (Table 7.2).

7.4.1 Laccases (Phenol Oxidase)

Laccase (EC.1.10.3.2) is a low molecular weight multi-copper oxidase family enzyme with less substrate specificity and is potent to degrade a wide range of xenobiotic compounds and aromatic and nonaromatic substrates. It has great importance in biotechnological approaches as it has bioremediation capacity and does not use readily available oxygen as an electron acceptor. Several low molecular weight compounds act as the efficient redox mediator in electron transfer steps of laccase reaction. It is able to degrade and decolorize the phenolic compound, aromatic azo compounds, etc. It oxidizes the aromatic amine using Cu^{2+} as the mediator. Majority of laccases are either fungal origin or plant origin, and very few have bacterial origin having great importance in biotechnological approaches (Kour et al. 2019b).

7.4.2 Lignin Peroxidase

Lignin peroxidase (Lip) (EC.1.11.1.14) is an extracellular glycosylated heme protein, topical owing to their high redox potential and prospective industrial applications. The potential applications of Lip span through sectors such as biorefinery, textile, energy, bioremediation, cosmetology, and dermatology industries. The huge number of potentials attributed to lignin peroxidase is occasioned by its versatility in the degradation of xenobiotics and compounds with both phenolic and nonphenolic constituents. Several iso-enzymes of Lip are produced; each of them is very specific and distinct in terms of pH, sensitivity, and substrate specificity.

Table 7.2 List of some industrially important microbial enzymes responsible for dye degradation along with its reaction specificity

Name of the Enzyme	Producing organisms	Dye degraded	% of degradation	Reference
Laccase	*Geobacillus stearothermophilus*	Indigo carmine, Congo red, Remazole, Brilliant Blue R, Brilliant Green	99, 98, and 60, respectively	Mehta et al. (2016)
	Micrococcus luteus	CI Acid Black 234, CI Acid Black 210	92.2, 96.4	Kanagaraj et al. (2015)
	Aeromonas sp. DH-6	Methyl orange	100 (in 12 h in optimum conditions)	Du et al. (2015)
	Providencia sp. SRS82	Acid Black 210, Triazodye	Maximum in 90 min under optimum condition	Agrawal et al. (2014)
Azoreductase	*Shewanella* sp. Strain IFN4	Reactive Black 5, Acid Red 88, Direct Red 81, Acid yellow 19, and Disperse Orange 3	Showing maximum degradation using Reactive Black 5 as a substrate	Imran et al. (2016)
	Pseudomonas entomophila BSI	Reactive Black 5	93 after 120 h of incubation	Khan and Abdul (2015)
	Aeromonas sp. DH-6	Methyl orange	100 (in 12 h in optimum conditions)	Du et al. (2015)
	Lysinibacillus sp.	Reactive orange 16	–	Bedekar et al. (2014)
	Enterobacter sp SXCR	Congo Red	Wide Range	–
	Alishewanella sp. strain KMK6	Reactive Blue 59	More than 90	Kolekar et al. (2013)
Peroxidase	*Pseudomonas putida MET94*	Anthraquinonic or azo dyes, phenolics, methoxylated aromatics	–	Santos et al. (2014)
	Providencia sp.SRS82	Acid Black 210	Maximum in 90 min under optimum condition	Agrawal et al. (2014)
Tyrosinase	*Providencia* sp.SRS82	Acid Black 210	Maximum in 90 min under optimum condition	Agrawal et al. (2014)

7.4.3 Manganese Peroxidase

Manganese peroxidase (MnP) (E.C. 1.11.1.13. Mn^{2+}: H_2O_2 oxidoreductases) belongs to the family of oxidoreductases, to specifically act on peroxide as acceptor (peroxidases), is an extracellular heme protein which catalyzes the H_2O_2-dependent oxidation of lignin derivative-based polymers. MnP is a specific enzyme that can oxidize Mn^{2+} to Mn^{3+}, which diffuses from the enzyme surface and in turn oxidizes the phenolic substrate, including lignin model compounds and some organic pollutants.

In nature, MnP catalyzes plant lignin depolymerization as component of ligninolytic enzyme complex. So it is one of the most common lignin degradation enzymes and has great and potential application in the field of agriculture for degradation of some cellulose, hemicellulose and lignin, etc. To protect the environment, it was widely used in many industrial fields for the degradation of some recalcitrant organic pollutants such as polycyclic aromatic hydrocarbons, chlorophenols, industrial dyes, and nitroaromatic compounds, which are very harmful to human health. Recently, more and more attention has been paid to the value of bioremediation of MnP enzyme. Previous research had been conducted extensively on azo dyes which could be efficiently degraded by the purified MnPs isolated from *P. chrysosporium*, *Lentinula edodes*, *Trametes versicolor*, *Dichomitus squalens*, *Stereum ostrea*, and *Irpex lacteus* (Yadav et al. 2018, 2019).

7.4.4 Tyrosinase

Tyrosinase (EC 1.14.18.1) is formally termed monophenol monooxygenase and is copper-containing enzyme. It is also known as phenolase, monophenol, oxidase, and cresolase. It is functionally similar to oxidoreductase enzyme.

7.4.5 Anthraquinone Reductase

Reductases are the enzymes which catalyze the reductive cleavage of sulfonated bond to produce colorless aromatic amine products. Several studies have been reported that bacterial cytoplasmic reductases have a suitable application for the purpose of environmental biotechnology.

7.5 Isolation and Identification of Dye Degrading Microorganisms

7.5.1 Sample Collection

Textile industrial units are using lots of water for dyeing process, and this untreated wastewater effluents are directly discharged into the canals. These effluents were collected in sterile containers which were rinsed with the same effluents and then collected, stored at low temperature (4 °C), and transported to the laboratory for analysis. The following process of laboratory analysis path way is used in isolation, decolorization and degradation (Fig. 7.1).

7.5.2 Physicochemical Properties

7.5.2.1 Temperature

Temperature is the most familiar ecological factor and is also easily measured. An ordinary mercury thermometer was used for recording the temperature at sample sites.

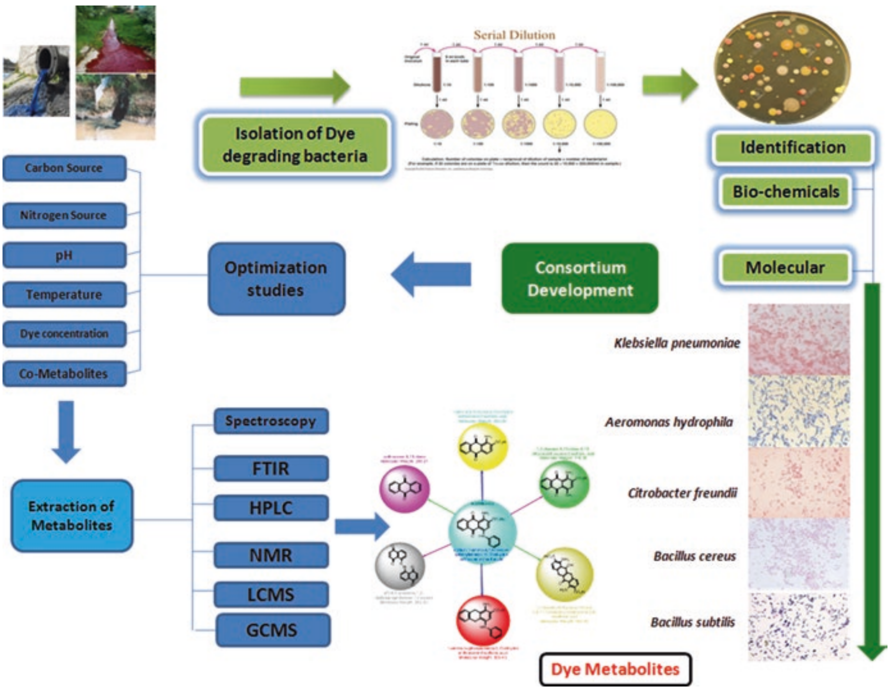


Fig. 7.1 Schematic representation of optimization of Acid Blue 25 degradation process and its metabolites analysis

7.5.2.2 pH

Collected effluents were used to test for pH with the help of calibrated pH meter.

7.5.2.3 Conductivity

Conductivity is a measure of a material's ability to conduct an electric current which is a numerical expression also applied for dye effluents. The ability of ions depends on the presence of total ions, their total concentration, mobility, valency, and relative concentrations including temperature. It has been measured by using a simple device "conductivity meter," and their values were expressed in "micro mho/cm."

7.5.2.4 Turbidity

Turbidity is the clarity of the effluent water. It is caused by dye silt and soil particles, therefore limiting the growth of an organism which is adapted for textile effluent conditions. A nephlometer was used to measure the turbidity of effluents. An appropriate amount of sample was placed in the test tubes, and the turbidity at different ranges was measured, and the values were recorded as "NTU" (nephelometric turbidity units).

7.5.2.5 Dissolved Oxygen

Dissolved oxygen levels in effluent water depend on the physical, chemical, and biochemical activities of microorganisms amended. The amount of dissolved oxygen is also resulted from photosynthetic activity of aquatic plants. Usually, the purification of water system depends on the presence of sufficient amount of oxygen in dissolved nature. In general, the effluent water met with organic pollutants that ultimately deprive the dissolved oxygen content.

7.5.2.6 Biochemical Oxygen Demand

The biochemical oxygen demand (BOD) is an empirical test determined by using dilution method in which the relative oxygen requirements are present in natural waters. It measures the oxygen required for the biochemical degradation of organic material and the oxygen used to oxidize inorganic materials, such as sulfates and ferrous ions.

7.5.2.7 Chemical Oxygen Demand

The chemical oxygen demand (COD) is a reflux method used as a measure of the oxygen equivalent of the organic matter content of water, that is, susceptible to oxidation by a strong chemical oxidant.

7.5.2.8 Total Dissolved Solids

Total dissolved solids (TDS) is denoted mainly by the acclimatization of various minerals present in water. TDS does not contain any gas and colloids. Perhaps, they can be determined by the residue left after evaporation of the filtered sample.

7.5.2.9 Total Suspended Solids

The suspended impurities present in the water mostly are of organic nature and pose severe problems of water pollution. The total suspended solids (TSS) is determined by the difference between total solids and total dissolved solids.

7.5.2.10 Phosphates

Phosphorus is available in nature only as phosphates. It is one of the major nutrients and is necessary for the production of nucleic acids, phospholipids, and variety of phosphorylated compounds. The main sources of phosphates are phosphorous-bearing rocks, discharge of domestic and industrial wastes, detergents, and agricultural runoff. Phosphates are mainly involved in biological productivity and enrichment of nutrients by ‘eutrophication.

7.5.2.11 Nitrates

Different forms of nitrogen are available in the nature which has greatest interest in soil or water, in order to facilitate oxidation state, nitrate, nitrite, ammonia, organic nitrogen. All these forms of nitrogen are biochemically interconvertible. Nitrate generally occurs in trace quantities in effluent water but may attain high levels of groundwater.

7.5.2.12 Chlorides

Chloride is a free ion, one of the major inorganic anions in water. The salty taste produced in potable water is due to the presence of chloride ions. In a neutral alkaline condition, slightly alkaline condition, the use of potassium chromate can indicate the presence of chloride.

7.5.3 Isolation of Dye Degrading Bacteria From Textile Effluents

Microorganism assimilates the nutrients as the basic requirements of growth. But there is diversity in the use of organic and inorganic compounds. Thus, culture media vary in composition depending on the microorganisms which are to be cultivated. Culture media also provide inorganic salts and may be supplemented with one or more organic diversified compounds for enrichment. The most commonly used medium for dye degradative bacteria is as follows:

Nutrient broth: peptone 5.0 g; beef extract 3.0 g pH 7.0; NaCl 2.5 g; and distilled water 1 L.

Glucose broth: peptone 10 g; glucose 5 g; NaCl 5 g; distilled water 1 L.

Zhou and Zimmermann medium: (g/l glucose 0.5%, yeast extract 0.5%, $(NH_4)_2SO_4$ 0.5 g/l, KH_2PO_4 2.66 g/l, Na_2HPO_4 4.32 g/l, distilled water 1 L. The above ingredients were mixed thoroughly, and its pH was adjusted and sterilized at 121 °C, at 15 lbs for 15 min.

7.5.3.1 Serial Dilution Method

A amount of 10 ml of effluent was suspended in 90 ml of sterile water to make a microbial suspension. Serial dilutions (10^{-2}–10^{-7}) were made. Finally, one ml of aliquot of various dilutions is placed in Petri dishes, by incorporating the sterile, cooled, and molten media; after 24 h of incubation, the culture plates were analyzed for bacterial colonies.

7.5.3.2 Secondary Screening

The selected bacterial isolates are then enriched in culture having Zimmerman media along with different dyes separately. All the strains were individually inoculated into the Zimmerman medium amended with 50 mg/L of each dye (Acid Blue 25, Reactive Blue 19, and Brilliant Blue R). The medium should be sterilized, and

10 ml of this medium is dispensed into different sterilized test tubes. The test tubes were inoculated with one loop full of selected organisms and incubated at 37 °C for 3 days under static and shaking conditions. The color change was noted at regular time intervals. The organisms that show maximum decolonization were selected and used for further biodegradation studies. Among the isolates, the selection of optimal organisms found who possess maximum decolonization of all three anthraquinone dyes.

7.5.4 Identification of Selected Bacterial Strains

7.5.4.1 Biochemical Methods

The "Bergey's Manual of Determinative Bacteriology" served as a practical guide for the identification of bacteria. The identification was based on the five different criteria: (i) energy and carbon sources, (ii) mode of location, (iii) morphology and gram's stain reaction, (iv) gaseous requirement, and (v) ability of spore formation. These five common features used for its identification and the information gathered for an unknown bacterium and their identification is as follows:

1. Morphological: shape, arrangement, capsule, gram's stain reaction, spore stain, and motility.
2. Cultural characters: nutritional state, colonies, temperature, growth, form, margin, elevation, and density.
3. Growth: surface growth, cloudy, sediment, etc.
4. Biochemical tests: The biochemical analysis is usually made to separate the microorganisms easy. The main biochemical aspect of the bacterial identification is the fermentations (glucose, sucrose, lactose, fructose, maltose, mannitol, etc.).

7.5.4.2 Molecular Identification

The selected potential strains were identified by comparing their 16S r DNA sequence with16S r RNA sequence data reference from the gene bank database using BLAST. The PCR products obtained from amplification with the universal primers and targeting r RNA gene was sequenced by using upstream and downstream primers. All the bacterial isolates and their sequences were confirmed by the using NCBI / BLAST. Program www.ncbi.nib.gov/blast. Phylogenetic analysis method was used for comparative analysis of genes in order to show the evolutionary relationships which is mainly performed by aligning the sequences by using ClustalW; then, the phylogenetic tree was constructed by MEGA 5 Software.

7.5.4.3 Scanning Electron Microscopy

The scanning electron microscope (SEM) analysis is mainly focused on the identification studies. The selected bacteria were fixed on the surface of glass piece with 2.5% W/V glutaraldehyde overnight. Fixed samples were dehydrated by successive treatment with 30, 50, and 75% (v/v) ethanol and finally with pure ethanol for 1 h. The surface of particles was examined using a scanning electron microscope.

7.5.5 Optimization of Decolorization Process

7.5.5.1 Decolorization Assay

In order to know the decolorization activity of different microorganisms, the percentage of decolorization was calculated by decolorized broth and compared to the corresponding control. After decolorization, the contents were centrifuged at 10,000 rpm for 10 min in a cooling centrifuge. The supernatant was measured at maximum wavelength of each dye selected. Decolorization was expressed in terms of percentage and was calculated by using the following formula::

$$\text{Decolorization}\left(\%\right) = \frac{\text{Initial absorbance} - \text{Final absorbance}}{\text{Initial absorbance}} \times 100$$

7.5.5.2 Development of Bacterial Consortia

After initial screening of the bacterial cultures with the dye decolorization ability, different potential bacterial strains were selected, namely *Klebsiella pneumonia, Aeromonas hydrophila, Citrobacter freundi, Bacillus cereus, and Bacillus Subtilis*. A loop full (0.01 ml) of each selected culture suspenctions were inoculated individually into Nutrient broth and incubated for 24 h to form consortium. 2% inoculum containing approximately 2×10^8 cells/ml in different combinations in 250 ml flask. The selected bacterial strains were used in different combinations to construct the different consortiums.

7.5.5.3 Optimization of Dye Decolorization

The decolorization of Acid Blue 25 textile anthraquinone dye by bacterial consortium was optimized with various parameters such as pH, temperature, carbon and nitrogen sources, and dye concentration for efficient dye decolorization.

The optimization of dye decolorization with developed consortium was analyzed by changing the different parameters like carbon source (glucose, sucrose, fructose, maltose, mannitol), different nitrogen source (yeast extract, beef extract, and peptone), different temperatures (25–45 °C), different PH (5–9), dye concentrations (100 mg to 500 mg/l), in Zimmermann's medium, and the flasks were incubated at 37 °C at static conditions.

7.5.5.4 Extraction of Biotransformed Products

In order to extract the biotransformed metabolites from the decolorized medium, the supernatant was collected, and the degradation products of Acid Blue 25 were extracted with ethyl acetate and concentrated under reduced pressure with the help of rotator evaporator and dried to powder form. Dried extract was dissolved in 10 ml of methanol. After solvent evaporation, collected residue is examined for Uv-Vis spectrophotometer, Fourier Transform Infra-Red Spectrophotometric, Liquid chromatography – Mass Spectrophotometer and Nuclear Magnetic Resonance.

7.5.6 Physicochemical Analysis of Effluents

The physico-chemical parameters of textile industrial effluents were analyzed. The pH ranged between 7.2 and 10.5, and the temperature was at 28–37 °C. Electoral conductivity showed minimum 2.42–6.31 values which is very toxic to plants and microorganisms due to high salt concentration; the total suspended solids of the effluents was observed at a range of 130–650 mg/L. Total dissolved solids is recoded high 8090 mg/L. The high range of TSS and TDS of industrial effluents shows effects on growth and metabolism of plants and microorganisms due to lack of soil permeability and aeration. There was a high load of BOD (500 mg/l), whcih indicate the low or zero levels of oxygen available. Biochemical oxygen demand and chemical oxygen demand showed high value 500 and 2548 mg/l which implies the toxicity by chemicals which indicates that the dissolved oxygen of the effluents was depleted due to cause of chemical pollution. Phosphates, nitrates, and chlorates showed 10.65, 247, and 2750 mg/ L, respectively, which are more than the permissible range of WHO. The chloride extent was 2750 mg/l, and a very high range was recorded in textile effluents.

Selected microbial strains were conformed for their decolorization efficiency of three dyes at 50 mg/l (Acid Blue 25, Reactive Blue 19, and Brilliant Blue R). Usually, maximum decolorization was showed by bacteria within 1 week of incubation. Screening experiments assessed the potential of selected bacterial strains for decolorization ability of anthraquinone dyes in liquid medium. The decolourization ability of the selected bacterial isolates were ranging from 11.32% to 92.38% (Fig. 7.2).

Five potential bacterial strains were labeled as B1, B2, B3, B4, and B5 according to their maximum decolorization efficiency. The selected bacterial strains were identified by using biochemical and molecular analysis (16S rRNA sequencing) and

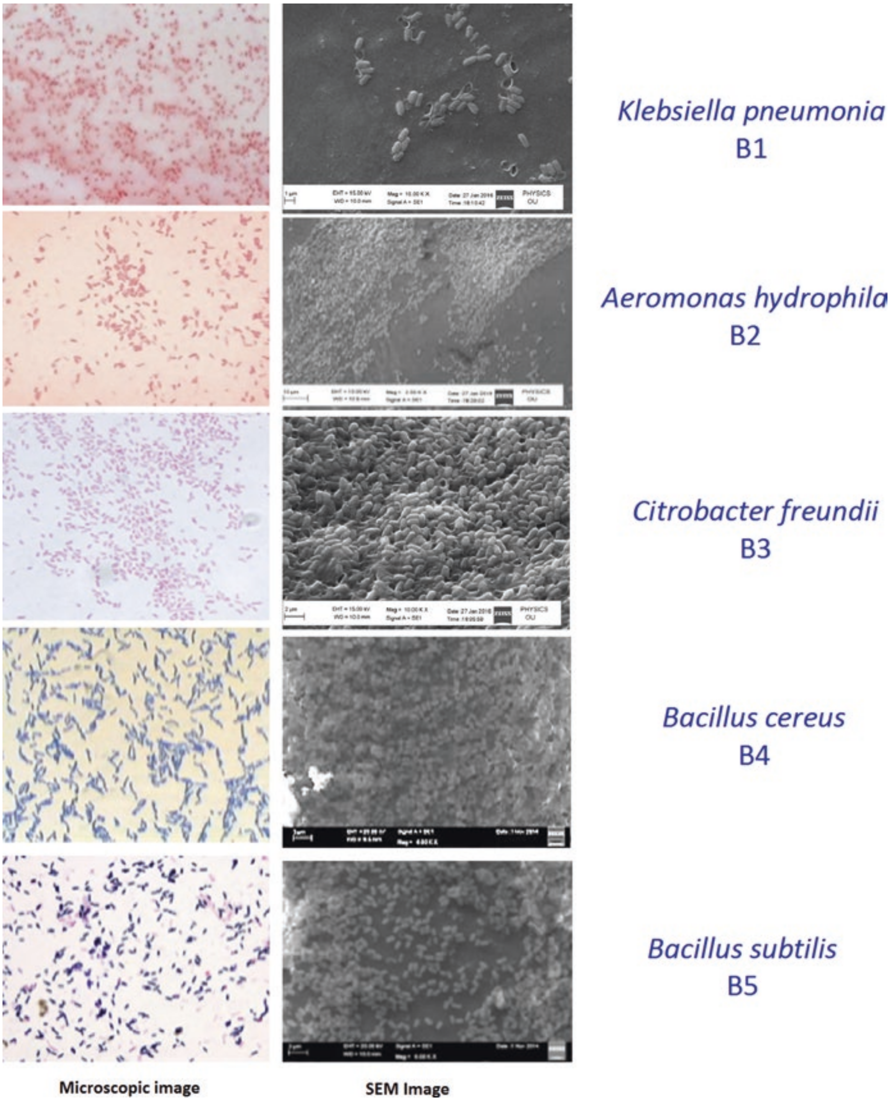


Fig. 7.2 Microscopic and SEM images of selected bacterial isolates

identified: B1 as *Klebsiella pneumonia*, B2 as *Aeromonas hydrophila,* B3 as *Citrobacter freundii,* B4 as *Bacillus cereus* and B5 as *Bacillus subtilis* (Table 7.3).

The maximum decolorization extent of Acid Blue 25 was observed with *Klebsiella pneumonia* (92.36%) within 48 h of incubation followed by *Aeromonas hydrophila* (84.48%), *Citrobacter freundii* (89.31%), *Bacillus cereus* (68.05%), and *Bacillus subtilis* (69.13%) within 96 h of incubation. From the present study, the *Klebsiella pneumonia* was proved to be highly potential bacteria for its

Table 7.3 Characterization of data of compounds A, B, C, D, E, and F forms during microbial degradation of Acid Blue 25

Compound	Name	Mass (m/z)	IR(cm^{-1})	^{1}H-NMR(δ)
A	(E)-4-((5,8-dihydroxy-5,8-dihydro naphthalen-1-yl) diazenyl)-5,8-dihydroxy-5,8-dihydronaphthalene-2-sulfonic acid	Calculated m/z = 430.43 [A]; Found: m/z=429[A-H]$^-$	3429(OH), 2924, 2854(Ar-H), 1457(N=N)	7.49–8.39 (m, 5H, Ar-H), 6.81 (s, 4H, C=C-H), 5.19 (dd, 4H, CH), 3.30 (s, 4H, OH), 2.24 (s, 1H, SO_3 of OH)
B	Sodium 1,4-diamino-10-hydroxy-9-oxo-9,10-dihydro anthracene-2-sulfonate	Calculated m/z = 342.30 [B]; Found: m/z=341[B-H]$^-$	3430(OH), 2925, 2853(Ar-H), 1654(C=O)	6.81–7.39 (m, 6H, Ar-H), 5.35 (s, 1H, OH), 4.02 (s, 1H, NH), 4.30 (s, 4H, NH_2).
C	5-amino-6-hydroxy-8-(phenylamino)-2,3-dihydronaphthalene-1,4-dione	Calculated m/z = 282.29 [C]; Found: m/z=281[C-H]$^-$	3428(OH), 2923, 2851(Ar-H), 1655(C=O)	7.15–8.29 (m, 5H, Ar-H), 5.32 (s, 1H, OH), 5.09 (s, 1H, CH), 4.29 (s, 4H, NH_2), 2.27–3.16 (m, 4H, CH_2).
D	Anthracene-9,10-dione	Calculated m/z = 208.21 [D]; Found: m/z=207[D-H]$^-$	2923, 2854(Ar-H), 1657(C=O)	7.85–8.29 (m, 8H, Ar-H).
E	3-hydroxy bicyclo[4.2.0] octa-1(6),2,4-triene-7,8-dione	Calculated m/z = 150.13 [E]; Found: m/z=149[E-H]$^-$	3427(OH), 2921, 2854(Ar-H), 1655(C=O)	6.87–7.39 (m, 3H, Ar-H), 5.33 (s, 1H, OH), 5.10 (s, 1H, CH), 3.79 (s, 1H, OH).
F	3-hydroxybicyclo[4.2.0] octa-1(6),2,4-triene-7,8-dione	Calculated m/z = 148.12 [F]; Found: m/z=147[F-H]$^-$	3429(OH), 2924, 2854(Ar-H), 1658(C=O)	7.17–7. 93 (m, 3H, Ar-H), 5.33 (s, 1H, OH).

The Acid Blue 25 dye appears to form six major compounds during microbial degradation. This is indicated in the LCMS analysis of the analyte; the major peaks of its LCMS (Fig. 7.1) are labeled as A, B, C, D, E, and F. These structures were characterized on the basis of negative-ion mass spectral analysis, IR, and 1H-NMR spectral analysis, and identified their names as (A)-(E)-4-((5,8-dihydroxy-5,8-dihydronaphthalen-1-yl)diazenyl)-5,8-dihydroxy-5,8-dihydro naphthalene-2-sulfonic acid, (B)-sodium 1,4-diamino-10-hydroxy-9-oxo-9,10-dihydroanthracene-2 sulphonate, (C)-5-amino-6-hydroxy-8-(phenylamino)-2,3-dihydronaphthalene-1,4-dione,(D)-anthracene-9,10-dione,(E)-3-hydroxybicyclo[4.2.0] octa-1(6),2,4-triene-7,8-dione and (F)-3-hydroxybicyclo[4.2.0]octa-1(6),2,4-triene-7,8-dione

decolorization efficiency for Acid Blue 25. The results confirmed that the five bacterial isolates can decolorize Acid Blue 25 anthraquinone dye with high decolorization efficacy.

Biodecolorization of textile effluents by using microorganisms is gaining importance as it is eco-friendly, cost-effective, and nontoxic and produces less sludge or sediment. The bioremediation has a wide range of applications and prospects, because it is mediated by suitable microbial communities which completely decolorize and degrade the contaminated dyes on site. The main advantage of these processes is initiated from the fact that it is an aerobic treatment process, where the toxic aromatic substances formed as intermediates are completely mineralized by bacteria. Previously, several scientific reports showed that microbial degradation of dyes used in industrial level and the released products were characterized effectively.

Optimization studies were conducted with various parameters influenced on decolorization activity of Acid Blue 25 dye. Among these, the dye concentration significantly affects the decolorization potential of bacterial consortium. It was very clear that the decolorization of Acid Blue 25 dye by all groups of bacterial consortium increased with the increase in dye concentration up to 100 mg/l; hence, the maximum decolorization was observed at 100 mg/l of dye concentration; then, the rate of decolorization was reduced while increasing dye concentration up to 600 mg/l (600 ppm). Consortium-1(C-1) showed maximum 96% at 100 ppm, but enthusiastically C-1 showed the decolorization at very high dye concentration also (500 ppm). Among the 11 consortia except C-1, other combination of consortia from C-2 to C-11 showed moderate to less decolorization potential. With the increase in dye concentration to 200 ppm, all of them showed high percentage of decolorization. But in the range of 300 mg/l dye concentration and above up to 500 mg/l only C-3, C-6 showed good decolorization, i.e., 72.36% and 66.24%. In general, C-1 and C-2 are more effective.

The effect of different carbon sources (glucose, fructose, mannitol, maltose, sucrose, and starch) on dye decolorization efficiency of consortium was monitored. Highest decolorization was observed with C-1 when glucose was used as sole carbon source, though this C-1 consortia showed significant decolorization with all the carbon sources. Except C-1, remaining all other combinations of consortiums are not up to the level. It is also highlighted that in the presence of mannitol and starch used as a co-substrates, the decolorization efficacy of C-1 was observed very less than the C-2 and C-6.

Similarly, the effect of different nitrogen sources (Yeast extract, Beef extract, and Peptone) has been used on decolorization efficiency with different consortia. However, the maximum decolorization was observed with all the consortia with all nitrogen sources. The maximum decolorization efficacy (86.34%) with C-1 observed in the presence of yeast extract used as a sole source of nitrogen. The remaining nitrogen sources does not have any influence in the decolorization process. Instead of yeast extract, incorporation of other nitrogen sources (Beef extract and Peptone) are resuming the low decolorization efficacy of other consortium groups (C2-C11).

Out of the 11 consortia, C-1 is found to be the better combination of mixture of different bacterial species that has the optimum compatibility with each other and stringently shares the co-metabolites for their emergency need. Similarly, other influencing factors like 0.1% glucose and 0.5% yeast extract showed almost 90% similar activity in terms of degradation and decolorization.

It was noticed that almost all the individual bacterial strains performed dye degradation activity at pH range from 7 to 8. In contrast to the individual strains optimized by various pH ranges, the consortium-1 recorded high decolorization (90.73%) activity at the same range of pH significantly. Very less decolorization activity observed at pH 9. Hence, it was concluded that the neutral pH supports the decolorization activity in the medium. Due to the presence of mild alkaline, pH range mediated the enzymatic activity involved in dye decolorization in addition to the cellular growth.

The temperature played an important role in decolorization efficiency of consortium, over a temperature range (25–45 °C). Consortium 1 showed maximum of 88% decolorization; remaining all the other bacterial consortium groups showed the less decolorization efficiency from 60% to 75% when the temperature increased to 37 °C. C-1 recorded 96%. But during the high temperature (45 °C), there is a significant decrease in decolorization of all bacterial consortiums.

Degradation is a complex process that involves the tremendous multiple steps, though there are varieties of bacteria, which can bring about dye degradation efficiency, and each of the species is very obligate and particular to the group of dyes. But the treatment of textile effluent cannot be the easy process where the effluent contains different groups of dyes. Under these circumstances, the constructed microbial consortia with different specification can perform the dye treatment process very efficiently, which is very difficult or not even possible for individual strain.

7.5.7 Monitoring of Biotransformed Compounds of Acid Blue 25

The bioremediation process is receiving increased attention across the globe as eco-friendly and cost-effective for the remediation of dye-contaminated effluents. Generally, anthraquinone dye degradation exhibits intermolecular difference and their recalcitrance to biodegradation due to their complex structures and xenobiotic nature and also is considered as mutagenic and carcinogenic.

After the degradation, the biodegradable products were observed at different stages, which can help to propose and establish in order to predict the detailed mechanism and degradation pathways. Different microorganisms have different degradation pathways for variety of dyes, depending on the structure strategy of microbial system that used for degradation. But even small structural differences can affect the decolorization process. The major mechanism behind the biodegradation of synthetic dyes in microbial systems is the biotransformation of enzymes (Li et al. 2016; Rastegari et al. 2020a, b; Zeng et al. 2001).

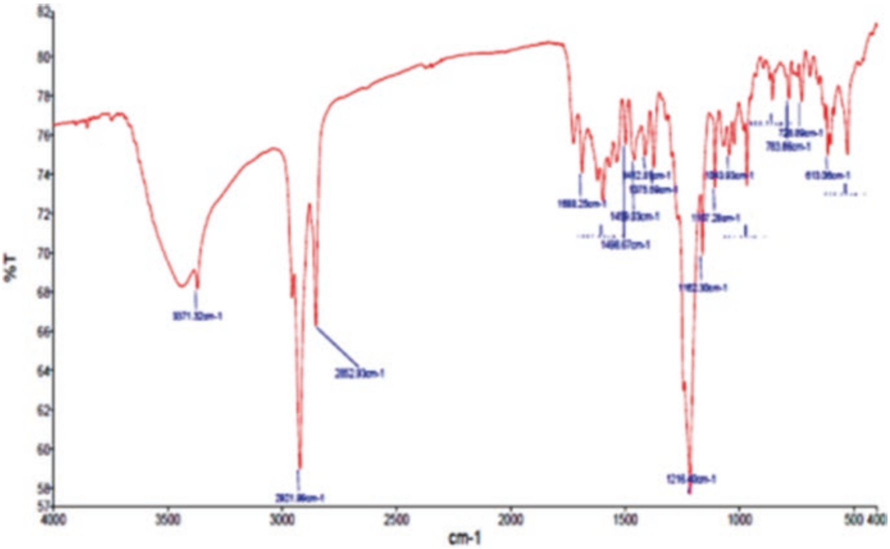
It is generally emphasized that reductases play an important role in bacterial dye decolorization. Among the synthetic dyes, anthraquinone dyes constitute second most important class of dyes with a wide variety of shades. The mechanism of biodegradation of anthraquinone dyes involves the reductive cleavage of C=O bond, C-N, and C-OH with the help of anthraquinone reductase enzyme under aerobic treatment process resulting to form colorless aromatic amines. In this regard, to explain the possible mechanism of dye decolorization, different analytical methods were used to identify the metabolites liberated from anthraquinone dyes after the bacterial treatment.

Usually, the chemical interaction of dye with the metabolic functioning state of cell was principally assessed by UV–Visible spectroscopy. The total disappearance of the absorption maxima after 48 h of Acid Blue 25 dye indicates the completion of decolorization process. Spectral analysis revealed a continuous decrease in the intensity of the original dye absorption peak 600 nm with significant decolorization efficiency in 72 h. After the chromophoric group cleavage, most of the intermediate metabolites seem to be trapped by nucleophilic or electrophilic reaction. This entire event is the crux of further pathway of degradation, where the metabolic burden may divert to activate accumulation or degradation.

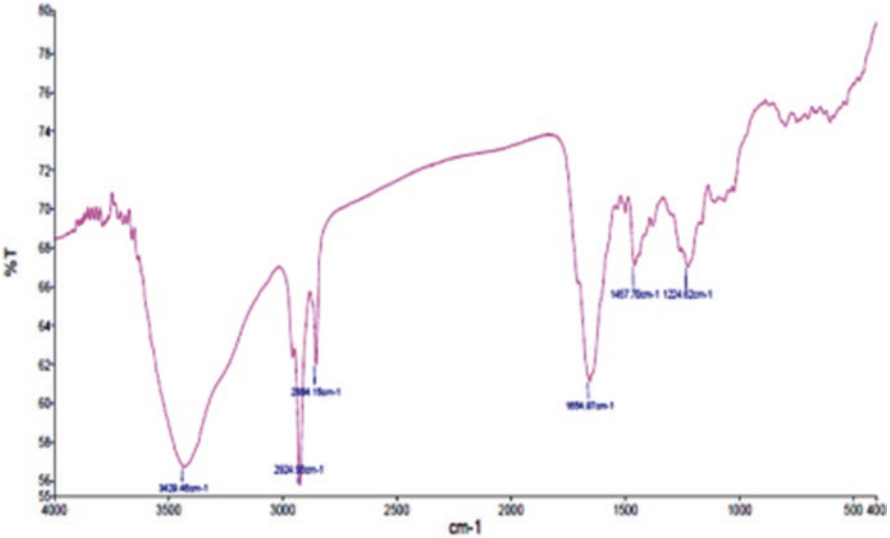
The structural analysis of the intermediates is established by FTIR, H NMR, and LC-MS. FTIR is a very useful technique to determine the optimal levels of factors and their interactions with each other. The anthraquinone dye, i.e., Acid Blue 25, degradation showed very interesting results by the facilitation of different and characteristic peaks. The FTIR spectrum of untreated dye (control) showed the specific peaks between 3371.32 cm^{-1} and 1043.93 cm^{-1} for the substituted benzene rings. The peak at 1162.30 cm^{-1} denotes C-N stretch amines, 1216.40 cm^{-1} with C-N stretch amines, 2852.93 cm^{-1} and 2921.99 cm^{-1} for O-H stretch carboxylic acids, and 3371.32 cm^{-1} for the O-H stretch phenol and also for the hydrogen bond with sulfonic acid. The FTIR spectra of treated dye sample showed at 1224.62 cm^{-1} for C-N stretch amines, an aliphatic amines peak at 1454.70 cm^{-1} of C-H bending alkanes, whereas 1654.97 cm^{-1} for C=C stretch alkenes, both 2854.15 cm^{-1} and 2954.93 cm^{-1} for O-H stretch carboxylic acids and 3429.46 cm^{-1} for O-H stretch phenols.

The peaks at O-H stretching of substituted benzene ring derivatives showed that the aromatic nature of the base dye compound treated with bacterial consortium was remained same. Usually the change observed that the surface participation of specific functional groups ensuring the peaks at 1216.40 cm^{-1}. A broad peak was observed at 3371.32 cm^{-1} of the biosorbent before treatment (control) which changed into sharp peak at 3429.46 cm^{-1} for the treated dye by the formation of C=O, O-H, carboxylic acid groups in the biotransformation of dye compounds (Fig. 7.3). In the FTIR spectra the Acid blue 25 degrades to 10 compounds, out of which 6-8 compounds are remaining. Another two compounds are C=C, O-H stretching aromatic structures with NH_2 and OH substitutes of sulfonic acids. At the same time, there are two more peaks remaining, but are reabsorbed or disappeared in the matter.

The H NMR spectrum of intermediates derived from the purified dye compounds during the degradation with retention time of 13 min (Fig. 7.4). The chemical shift region of the acid protons for the compounds 1 and 2 showed double singlets (one proton each) located at 12.343 pm and 13.850 pm. Typical ring aromatic protons of Acid Blue 25 skeleton peaks were observed from 7.20 to 8.40 pm. At the same time, the singlet is recorded at 7.80 pm which corresponds to aromatic ring carbon protons. There are two –OH protons and two –NH_2 protons, observed at 5.10 pm to



Acid Blue25 control



Test sample of Bacterial consortium

Fig. 7.3 FTIR spectrum of Acid Blue 25 before and after degradation

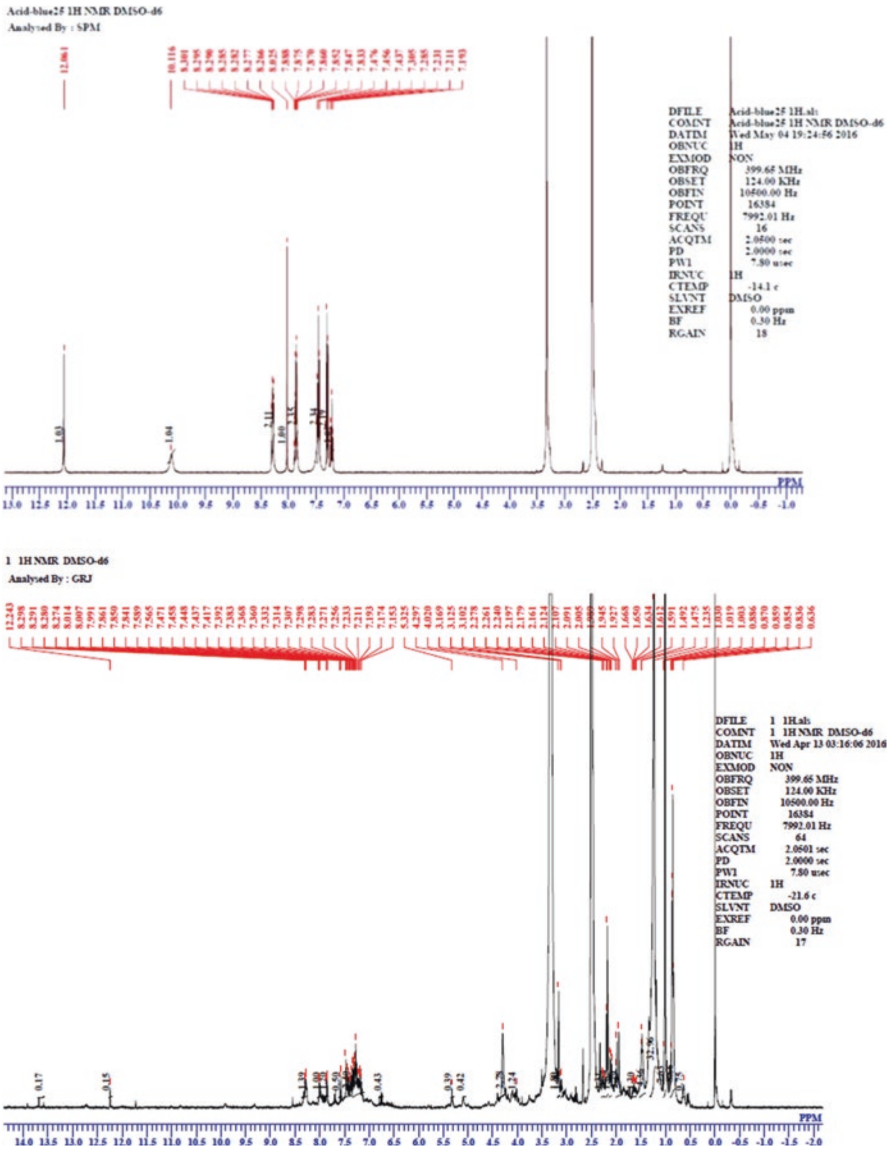


Fig. 7.4 NMR analysis of Acid Blue 25 degradation by Bacterial consortia before and treatment

5.35 pm and 4.00 pm to 4.30 pm, respectively. There are numerous broad peaks observed, and they are confirmed as solvent peaks which are similar to TMS reference line (i.e., baseline) of NMR spectra. No aliphatic signals were observed means no transformation of aliphatic compound produced. These observations are always monitored by correlating with controls.

Similarly, the eluted compounds were extracted and purified, and their biotransformed components were analyzed by LC-MS. The chromatogram was with four retention times (15, 1.995, 6.573, and 4.120), and their molecular mass was observed. The products after degradation mass spectrum peak at retention time of 4.103 showed a significant molecular mass ratios of 207 which corresponds to anthracene-9, 10-dione. No perfect and prominent peaks were at retention time of 3.113. The significant peaks with molecular mass details were recorded at the retention of 19.813, the molecular mass of 208 (M-H) corresponds to anthracene-9, 10-dione, and the molecular mass 303 (M-Na) gives 1-amino-9, 10-dioxo-9, 10-di hydroanthracene-2-sulfonic acid. Similarly at the retention time of 18.263, the peaks observed with the molecular mass of corresponding structures viz., anthracene-9, 10-dione & 1-amino-9, 10-dioxo-9, 10-di hydroanthracene-2-sulfonic acid are showed with 207and 281are same with the retention time of 4.103. At the retention time of 19.813, the molecular mass of the compound 342.30 (M-Na) corresponding to structure 5, 5-diazene-1, 2-diyl) bis (naphthalene 1, 4-dione) was derived after the dimerization of compounds was formed. Similarly, at the same retention time, molecular mass of 366.43 (M-Na) of a compound, also formed from the dimerization of phenylamine with 9,10 dihydroxy anthracene-2-sulfonic acid, corresponds to the structure 5 (Fig. 7.5) of (1-amino-4-(phenylamino)-9,10-dihydro-anthracene-2-sulfonic acid), and it was also confirmed from HNMR with negative mode of ionization.

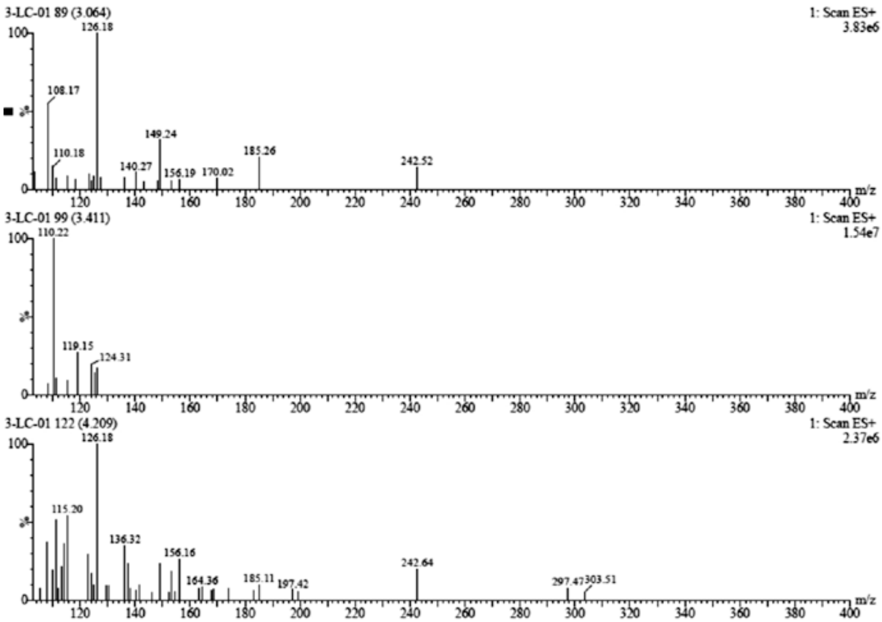


Fig. 7.5 (LCMS) Chromatogram and mass spectra of Acid Blue 25 by bacterial consortium

The compound 4 derived from the degradation of Acid Blue 25 by bacterial consortium showed the molecular mass of 452.46 at 18.113 retention time. It is corresponding to the structure formed from the dimerization of the liberated components; its corresponding structure is 1, 7, diamino-6-hydroxyl-12-oxo-5, 6, 11, 12-tetrahydrotetracene-2, 8-disulfonic acid. The compounds 2 and 3 are synthesized by splitting of phenylamine from the base compound. Their mass values were 303.29 and 318.30, and their corresponding structures were 1-amino-9, 10-dioxo-9, 10-dihdro anthracne-2-sulfonic acid and 1, 4-diamino-9, 10-dioxo-9, 10-dihydroanthracne-2-sulfonic acid. Acid Blue 25 forms abundant deprotonated molecule (M-H) (M-Na) upon ionization via negative-ion ESI. All deprotonated acid dyes fragment upon Collision Activated Dissociation (CAD) fragment ion loss of SO_2 and minor fragment ion loss of the secondary aromatic amine (Fig. 7.6).

The molecular mass value is the monoisotopic mass deprotonated molecule; sodium (Na^+) cation under negative ion ESI conditions will not be added to the m/z value. The m/z value of deprotonated molecule will differ by one or two Na^+ ions and/or one hydrogen weight of the dye. It is suggested that the lowest energy of fragmentation is the rearrangement of the sulfonate group to lose SO_2 as compared to the loss of secondary aromatic amine. Therefore, fragmentation pathway is very useful in determining possible structural substituents present on the anthraquinone backbone. The catabolic production of degraded components was identified; some of them are absorbed and reinteracted with derivatives resulted to form dimerized compounds (Table 7.3 and Fig. 7.6). However, these carbonyls, phenol oxides, and amino derivatives were transformed under static conditions during the biodegradation process. Very less information is available on molecular mechanism of transformation studies, particularly on anthraquinone Acid Blue 25 dyes. However, the biotransformation studies on anthraquinone Acid Blue 25 dye knowledge are very important not only for the development of efficient bioremediation, but also in search of harmless dyeing compounds being synthesized during degradation.

7.5.8 Toxicological Studies

The assessment of the ecological and genetic impact of dyes and their metabolites on agriculturally important microorganisms and plant populations is of great importance. Microbial toxicity was studied on *Pseudomonas aeruginosa*, and zone of inhibition was observed with undegraded Acid Blue-25, whereas its biodegradation products did not show growth inhibition. The dye Acid Blue-25 was found to be toxic, whereas the dye metabolites found to be stimulatory to the microorganisms.

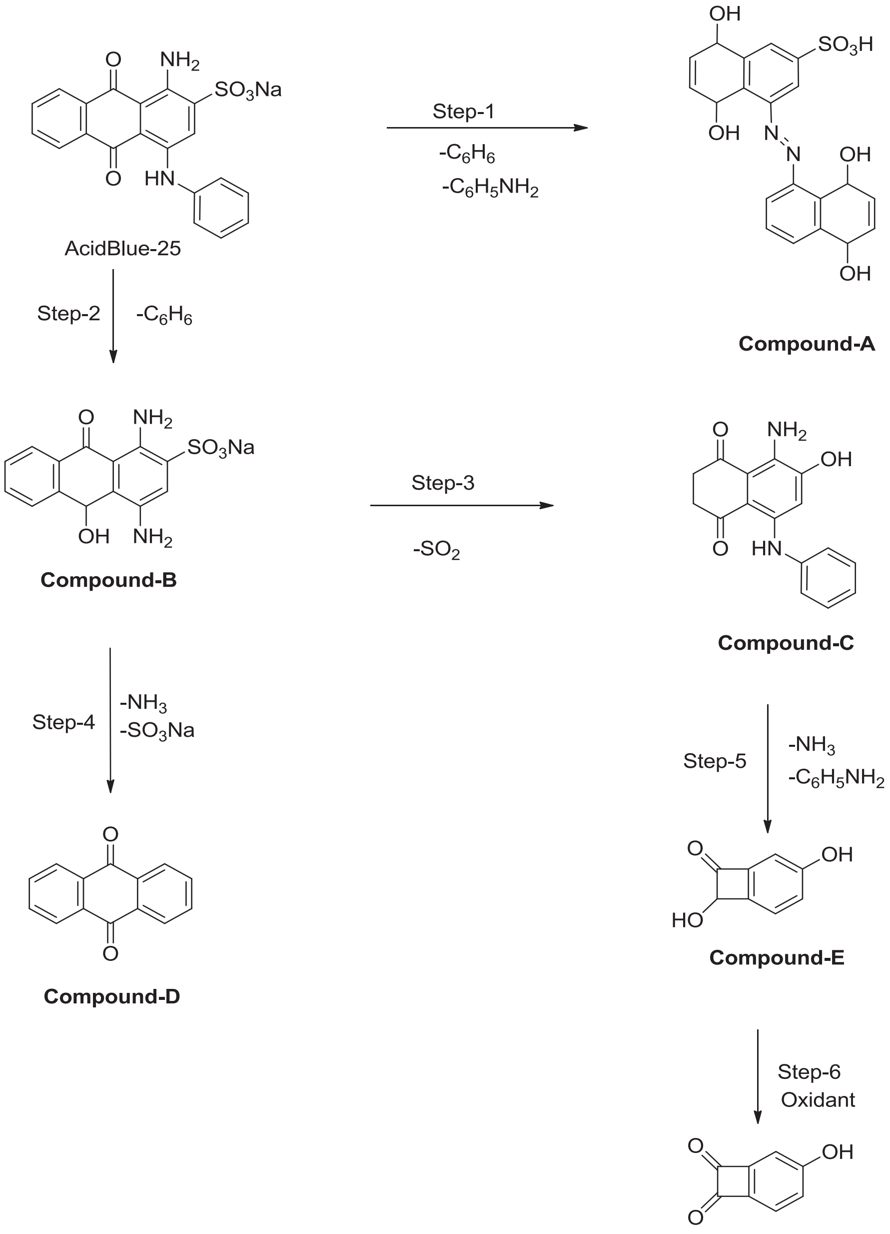


Fig. 7.6 Biodegradation pathway (structures and pathway of metabolites released during the decolorization) of Anthraquinone dye (Acid Blue 25)

7.5.9 Phytotoxicity Studies

After the degradation process the bio-transformed metabolites released were tested for toxicity analysis which is helpful for the safe discharge of effluents from the textile industry. There are two studies in practice, phytoremediation and *microbial remediation. Phaseolus mungo* seeds were used to evaluate the phytotoxic effects of Acid Blue 25 and its degraded products. Acid Blue 25 dye and its degraded products in different concentrations (100–10,000 ppm) were along with the appropriate controls. Germination (%), plumule, and radicle lengths are considered as parameters to verify the toxicity levels of metabolites in the biotreated effluents. It was proved that the extracted degradation products are nontoxic in nature, resulting in good germination as well as plumule and radicle length of *Phaseolus mungo* when compared to the untreated dyes.

7.6 Conclusions

From the above information summarized and for growing evidences from literature, and of the work performed by researchers, the following conclusions were made.

1. Bioremediation is a cost-effective, better environmental-friendly, and alternative method for the color removal from the textile industry effluents.
2. Biological treatment has been effective in reducing the textile industry effluents, which has lower operating cost than other remediation systems used for decolorization process.
3. Microbial consortium has become a very good source for the textile industries, in getting rid of problems by biodegradation and decolorization. Because of their low cost and efficiency, they are highly recommended for the industries by making use of consortium for the proper disposal of textile effluents.
4. For the technology transformation there is a need to design a mini reactor or bio filtration unit for the better treatment of textile effluents before they discharge in to the environments showed significant decolorization recorded with bacterial consortium. Maintenance and applicability of this bioreactor were very simple and easy-to-use onsite for small-scale industries.
5. In short, the main advantages of bacterial decolorization process are as follows:
   (a) Bacterial treatment uses no chemicals.
   (b) Bacterial consortia dispose no harmful matter into the environment, and the process eliminates chemical processing and unusual vapor.
   (c) No energy is required; consequently, it is zero-carbon process; no complicated technology is required.
   (d) Therefore, we conclude that using bacterial consortia is the most effective way of treating industry effluents

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